UNIT-1

1. Explain the purpose of semantic elements in HTML5. Provide examples of three different semantic elements and their use cases.
2. Describe the Web Storage API. Differentiate between localStorage and sessionStorage with examples.
3. What are HTTP status codes? Provide examples of the following categories: 2xx (Success), 4xx (Client Errors), and 5xx (Server Errors).
4. Explain the CSS3 box model. Draw a diagram illustrating the model and describe the purpose of each component (content, padding, border, margin).
5. Differentiate between CSS3 Grid and Flexbox. Provide a scenario where Grid would be preferred over Flexbox, and vice versa.
6. Explain how to create a responsive web design using Media Queries in CSS3. Provide an example of a media query that targets mobile devices.
7. Describe the purpose of the viewport meta tag in responsive web design. How does it affect the rendering of web pages on mobile devices?
8. Explain CSS3 transitions and animations. Provide an example of a CSS animation that moves an element across the screen.
9. What is Tailwind CSS? Describe its utility-first approach and explain how it differs from traditional CSS frameworks.
10. Using Tailwind CSS, write the necessary class names to create a button with a blue background, white text, and rounded corners.
11. Explain the concept of pseudo-classes in CSS. Provide 3 examples. (Context outside: Explain the difference between pseudo-classes and pseudo-elements.)
12. Describe the concept of CSS specificity. Why is it important and how does it affect styles applied to elements?
13. Explain the use of the calc() function in CSS. Provide an example of how it can be used to dynamically size elements.
14. What is the purpose of the object-fit property in CSS? Provide examples of its different values and their effects.
15. Explain the concept of CSS variables (custom properties). How can they be used to improve maintainability and reusability of styles?

UNIT-2

1. Introduce JavaScript and explain its role in web development. Differentiate between client-side and server-side JavaScript.
2. Describe the different data types in JavaScript (primitive and non-primitive). Provide examples of each type.
3. Explain how to define and call functions in JavaScript. Provide an example of a function that takes parameters and returns a value.
4. Describe how to create and manipulate arrays and objects in JavaScript. Provide examples of common array and object methods.
5. Explain regular expressions in JavaScript. Provide an example of a regular expression that validates an email address.

6. Explain the purpose of let and const keywords in JavaScript. How do they differ from var?
7. Describe arrow functions in JavaScript. Provide an example and explain their advantages over traditional functions.
8. Explain destructuring assignment in JavaScript. Provide examples of destructuring arrays and objects.
9. Describe the spread and rest operators in JavaScript. Provide examples of their use cases.
10. Explain prototypal inheritance in JavaScript. How does it differ from classical inheritance?
11. What is a closure in JavaScript? Provide an example of a closure and explain its use case.
12. Explain callbacks in JavaScript. Why are they important for asynchronous operations?
13. Describe promises in JavaScript. How do they handle asynchronous operations? Provide an example of creating and using a promise.
14. Explain the async/await syntax in JavaScript. How does it simplify asynchronous programming?
15. Explain the "this" keyword in Javascript. How does "this" behave in different contexts? (Context outside: Explain how ".bind", ".call", and ".apply" change the context of "this".)

UNIT-3

1. What is MERN stack? Describe the role of each component (MongoDB, Express.js, React, Node.js).
2. Explain the concept of a serverless "Hello World" application in the context of MERN.
3. Describe the steps involved in setting up a basic Node.js server.
4. Introduce Node.js and explain its event-driven, non-blocking I/O model.
5. Explain the Node.js REPL (Read-Eval-Print Loop). How can it be used for development and debugging?
6. Describe the events module in Node.js. Provide an example of creating and emitting custom events.
7. Explain the os module in Node.js. How can it be used to access operating system information?
8. Describe the http module in Node.js. Provide an example of creating a simple HTTP server.
9. Explain file I/O operations in Node.js using the fs module. Provide examples of reading and writing files.
10. Describe how to access environment variables in Node.js. Why is this important for configuration?
11. Explain the purpose of the dotenv module in Node.js. How does it help manage environment variables?
12. Describe the process of creating and using custom Node.js modules. Provide an example.
13. Explain the concept of middleware in Express.js. Provide an example of a middleware function.
14. Describe how to handle HTTP requests (GET, POST, PUT, DELETE) in Node.js using Express.js.
15. What is the purpose of package.json in Node.js? How does it manage dependencies? (Context outside: Explain the semantic versioning used in package.json.)